Assignment 2: Systemic Games with Emergent Systems
Course: GAME 10017 Game Systems Design
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(I am so sorry the Canva pictures of my charts made the spacing of this document look so ugly, I tried using page breaker lines but it is still not good)

Q1. Justify Objects, Properties, Behaviours, and Relationships

For each mechanic, justify why you designed these specific objects, properties, behaviours, and relationships.

1.0 Assumptions

This design assumes a grid-based environment where the player can move freely and interact with objects. Basic movement and interaction function as foundational game mechanics and serve as inputs into the larger system.

1.1 Electric Battery

Object: Electric Battery

Permutations	Empty	Half-Full	Full-Charge
Battery Health			
Properties	• Charge State: Empty, Half-Full, Full-Charge • Visual: Colour coded meter (Red = Empty, Yellow = Half, Green = Full) • Capacity: Full (Green) • Drain/Charge Rate: 5 seconds = half-battery life		
Behaviours	• Stores electrical energy. • Drains energy when powering connected objects • Recharges when connected to a power source • Changes state based on current charge level		
Game State Feedback	Visual: • Colour: ◦ Charge meter visibly fills or empties. ◦ Colour changes based on charge state (Red, Yellow, Green) Audio: • Sound Effects: ◦ Buzzing while plugged into plug/door (transfering energy) ◦ Power down sound when emptied ◦ “Blip” noise when fully charged		
Systemic Feedback	• Battery state directly affects Powered Door Lock • Charge level influences puzzle outcomes or progression		

The Electric Battery is designed as a **state-based energy storage mechanism** within the system. It receives charge from the Plug and stores it until it reaches defined charge states (Empty, Half-Charged, Full-Charged). These states are clearly communicated through colour (Red, Yellow, Green), creating a **dynamic information structure** that allows the player to understand the current game state.

Its core **parameter** is charge value. This parameter establishes a **system threshold** at Full-Charge. The Battery directly alters the behaviour of the Powered Door. Through this dependency, it forms **coupled interactions** within the energy subsystem, where changes in one object influence another.

1.2 Powered Door

Object: Powered Door Lock

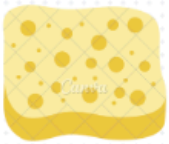
Permutations	Empty	Half-Full	Full-Charge
Power State			
Properties	• Power Dependency: Requires Electric Battery charge to function • Power Threshold: Full charge level required to unlock • Lock State: Locked or Unlocked • Default State: Locked when no power is supplied		
Behaviours	• Unlocks automatically when connected battery reaches required charge threshold. • Updates lock state based on battery state changes • Remains locked when connected battery is Empty		
Game State Feedback	Visual: • Power icon colour reflects battery state (Red = inactive, Yellow = partial, Green = active) • Lock icon visibly switches between locked and unlocked Audio: • Mechanical unlock sound when sufficient power is supplied		
Systemic Feedback	• Door state is directly affected by Electric Battery state • Creates a clear cause-and-effect loop between energy management and progression		

The Powered Door acts as a **gatekeeping mechanism** within the energy subsystem. It does not generate energy but stores it instead. It responds directly to the Battery's charge state. Its binary lock state (Locked/Unlocked) interacts with the Battery's parameter (charge level), producing a clear threshold-based transition.

When the door reaches Full-Charge and drains a battery, a quantitative change results in a qualitative environmental shift the door unlocks. This demonstrates **emergence** as progression is not caused by a single action, but by the structured relationship between connected objects. The door reinforces a **positive feedback loop**, since reaching the required threshold rewards the player with access to a new area and advances the overall game state.

1.3 The Sponge

Object: Sponge

Permutations	Empty	1 Water Tile	2 Water Tiles	3 Water Tiles
Absorbancy State				
Properties	• Water Hold State: Four states; Empty (0), 1 Tile, 2 Tiles, Full (3 Tiles) • Capacity: Maximum of 3 water tiles • Visual: Colour changes based on current water level • Can squeeze out stored water onto empty tiles			
Behaviours	• Absorbs water when moved onto a water tile • Increases water count up to maximum capacity (3) • Changes colour as water level increases • Deposits water when squeezed out on a empty tile • Decreases water count when water is released • Cannot absorb additional water when at full capacity			

Game State Feedback	Visual:
	• Colour
	  ◦ Colour change reflects water level (Blue → Green → Yellow → Red)
	• Animation
	  ◦ Sponge squeezes inwards when player releases water or absorbs water
	Audio:
	• Sound effects:
	  ◦ Absorb sound when collecting water
	  ◦ Squeeze/drip sound when releasing water
	  ◦ Subtle "full" cue when capacity is reached			

Systemic Feedback	• Acts as a transportion object between tiles • Forces players to manage what tiles the go on while pushing the Sponge • Sponge state affects puzzle outcomes			

The Sponge operates as a **state-based environmental control mechanism** within a separate subsystem. Its role is to absorb water tiles in order to clear pathways. Its main **parameter** is capacity (maximum of three water tiles), which directly defines its limits and shapes the system's **possibility space**.

Through the loop of absorb → hold → deposit water, the Sponge creates **context-dependent interactions** with the level layout. Player movement functions as the primary **game mechanic**, and each absorbed tile/deposited tile visibly changes the game state. The colour shift of the Sponge introduces a **dynamic information structure**, making its internal state readable at all times. Although the rules are simple, complexity emerges from how capacity, water placement, and spatial constraints interact over time.

1.4 Water

Object: Water

Permutations	Wet	Dry
Tile State		
Properties	• Environmental tile (water) • State: Present (Wet) or Removed (dry) • Volume Value: Single unit per tile (counts as 1 water) • Interactable: Can be collected by Sponge object	
Behaviours	• Remains static until interacted with • Transfers 1 unit of water into the Sponge when the Sponge moves onto it (if sponge not full) • Becomes Removed (dry) after being collected • Can be restored/created when Sponge deposits water onto a valid tile (if your rules allow it) • Cannot be collected if the Sponge is at full capacity	
Game State Feedback	Visual • Clearly readable water texture/colour on the tile • When collected: tile changes to an "empty/dry" version • When placed/restored: tile changes back to water version Audio • Splash/collect sound when absorbed • Drip/pour sound when placed/restored • Optional subtle ambient water sound when present	
Systemic Feedback	• Controls pacing/route planning (players must navigate) • Creates strategic constraints through limited supply + sponge capacity	

Water functions as an **environmental element** within the subsystem. It exists as tiles that occupy space and obstruct movement until removed. Its core **parameter** is presence (present/wet or removed/dry tile). When absorbed, a visible shift in the **game state** occurs, changing available movement/pathway options. Water does not exhibit behaviour independently, it participates in **coupled interactions** when acted upon by the player through the Sponge.

Although water follows rules, **complexity emerges** from the placement and distribution of water tiles within the level. The spatial arrangement determines which paths are accessible, shaping the player's decision making. This supports an elegant design approach, where a simple object with one defining property generates multiple potential configurations and outcomes.

1.5 Plug

Object: Plug

Permutations	
N/A	
Properties	• Power Output: Infinite/continuous energy source • Connection Type: Accepts Electric Battery object • Activation State: Active when battery is connected • Charge Rate: 5 seconds + half-battery life • Stationary Object: Fixed location within level
Behaviours	• Transfers energy to the Battery when connected • Increases battery charge over time (up to maximum capacity) • Stops transferring energy when battery is removed or fully charged • Does not store energy itself • Cannot function without a battery connected
Game State Feedback	Visual: • Animation ○ Plug, Plugging in and out Audio: • Noise of plug, unplugging • Noise of plug, plugging
Systemic Feedback	• Acts as the origin point of the energy loop. • Enables battery state changes which directly affect Door Lock state. • Creates spatial puzzle tension (player must transport battery to plug). • Forms a dependency chain: Plug → Battery → Door progression.

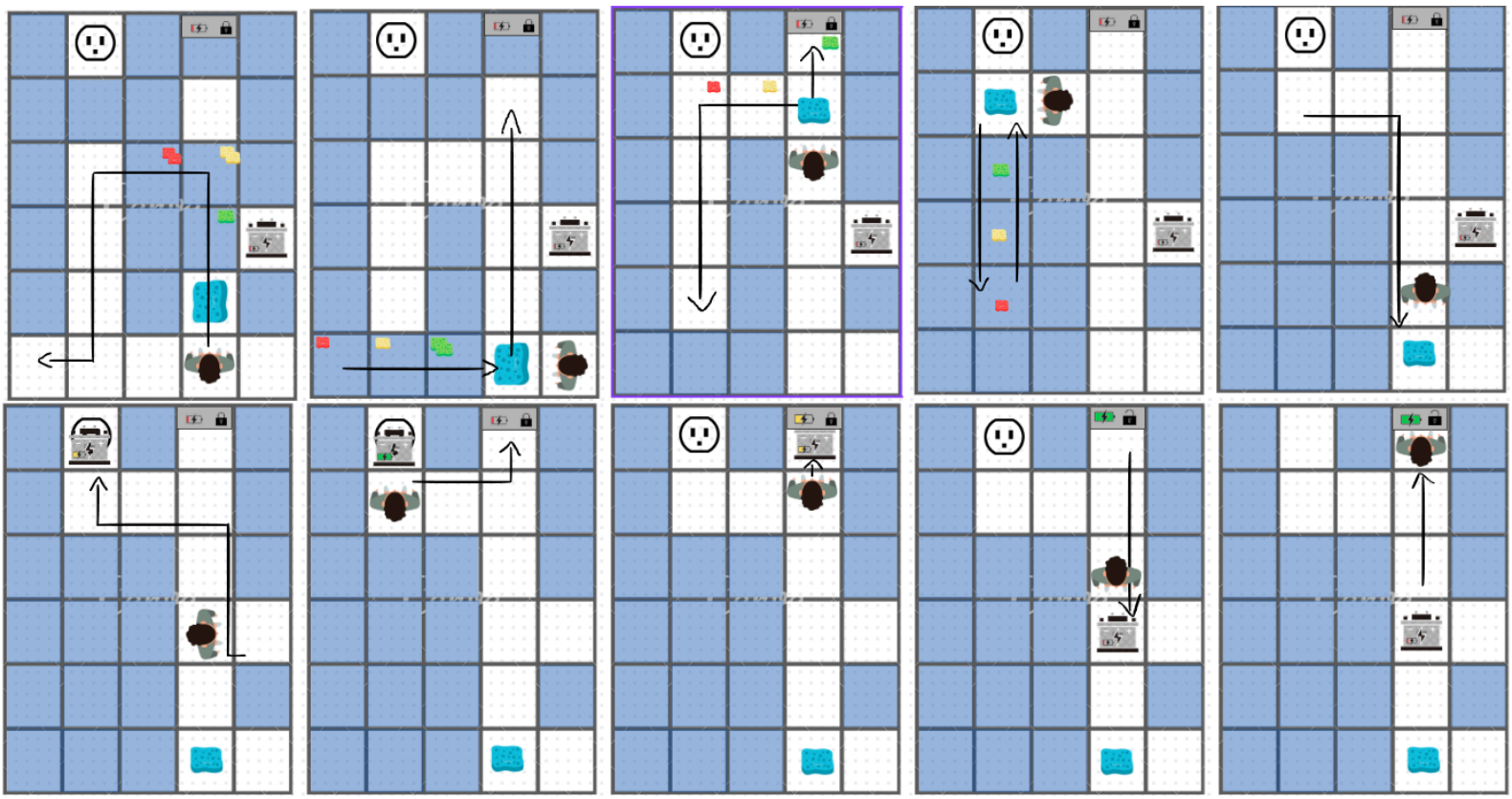
The Plug functions as the **energy source mechanism** within the energy subsystem. It does not necessarily store energy but transfers a charge to the Battery when connected. Its core **parameter** is charge rate, introducing time as a measurable factor within the system.

The Plug does not directly affect the environment, other than placement, and it forms a **coupled interaction** with the Battery. When connected, it increases the Battery's charge state over time, eventually enabling threshold-based state changes in the Powered Door. This creates a structured dependency chain within the subsystem.

Q2. Justify Emergence, Gameplay Possibility Space

Overall, justify how you used a minimal number of objects with a sufficient number of properties to generate a wide possibility space.

2.1 Scenario One

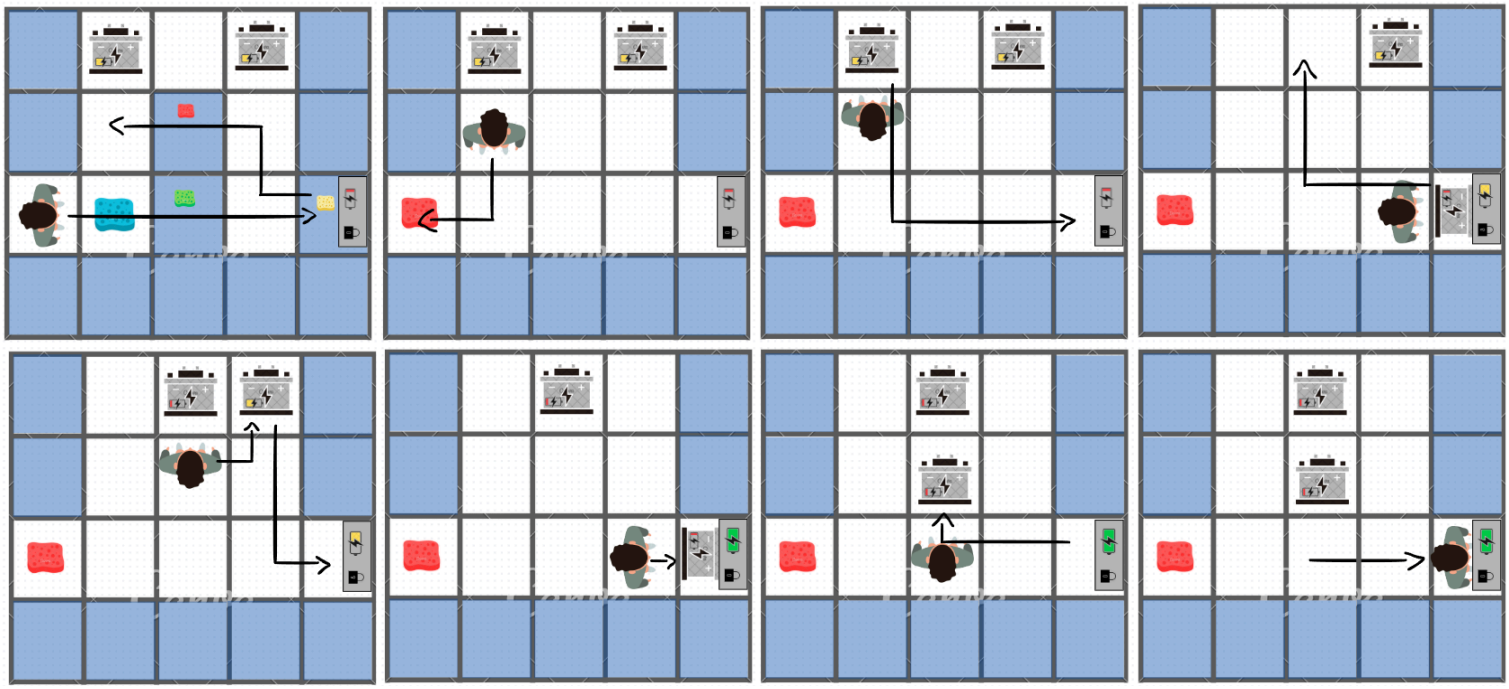


This design demonstrates **elegant complexity** by using a minimal set of objects (Water, Sponge, Battery, Plug, Door), each defined by clear parameters, to generate a wide **possibility space**. Individually, each object follows rules, but when combined they form a system of **coupled interactions**.

In Scenario One, the player coordinates spatial control and energy progression. The Sponge's limited capacity affects movement options, while the Battery and Plug introduce a charge parameter and **threshold-based** unlocking. A quantitative increase in charge results in a qualitative shift when the door opens.

The possibility space emerges from how these parameters interact. Different orders of clearing water or charging the Battery produce different game states within the same rule structure. This reflects the principle of **minimize to maximize**, where a small number of defined properties generate meaningful variation and emergent play.

2.2 Scenario Two



Scenario Two demonstrates **elegant complexity** as well using an even smaller set of objects. The player first uses the Sponge to clear the middle space, with its limited capacity shaping the **possibility space**. Unlike Scenario One, the player begins with two half-charged Batteries and no Plug.

The door responds to the combined charge of multiple Batteries, reinforcing its role as a **threshold-based mechanism**. Because the Batteries can be used in any order, the system supports **context-dependent interactions** without adding new mechanics. Emergence arises from coordinating spatial control and distributed energy resources, creating meaningful play through structured but flexible interactions.